

Name: _____

Class: _____

CHAPTER TEST

Chapter 10 Molecular manufacturing

Time permitted: 50 minutes

	Section	Number of questions	Marks available	Marks achieved
A	Multiple choice	15	15	
B	Short answer	5	15	
	Total	20	30	

Grade: _____

Scale:

A+	29–30	A	26–28	B	23–25	C	19–22	D	15–18	E	9–14	UG	0–8
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Comments:

Section A Multiple choice (15 marks)

Section A consists of 15 questions, each worth one mark. Each question has only one correct answer. Circle the correct answer. Attempt all questions. Marks will not be deducted for incorrect answers. You are advised to spend no more than 15 minutes on this section.

1 Which one of the following is not required for a successful collision between reactants?

- A Orientation
- B Energy
- C Angle of collision
- D Concentration

2 Stereoisomers are:

- A structurally rearranged molecules.
- B mirror image isomers.
- C identical isomers.
- D only found in amino acids.

3 To what does SSPS refer?

- A Peptide synthesis
- B Protein synthesis only
- C A method of growing proteins
- D Genetic modification of bacteria

4 Nanoparticles are:

- A atoms and molecules.
- B under 100×10^{-9} m in at least one dimension.
- C less than a nanometer.
- D have very small surface areas.

5 Mechanochemistry in molecular manufacturing:

- A manipulates the reaction rate of a reaction.
- B only occurs naturally.
- C is also known as the orientation effect.
- D produces proteins only.

6 Protective groups in molecular manufacture:

- A prevent non-desired products being formed.
- B are large functional groups.
- C are reactive sites on a molecule.
- D are esters and amides.

- 7 Polypeptides are:
- A the polymers of amino acids.
 - B peptide polymers.
 - C protein monomers.
 - D only found in nature.
- 8 Genetically modified bacteria:
- A produce all synthesised proteins.
 - B produce insulin amongst other useful medications.
 - C are dangerous.
 - D are found naturally in nature.
- 9 Which of the following is not found in DNA?
- A Adenine
 - B Cytosine
 - C Lysine
 - D Guanine
- 10 What is DNA's double-helix structure held together by?
- A Hydrogen bonds between bases
 - B Dispersion forces along the DNA strands
 - C Attraction between adenine and guanine
 - D Base pair covalent bonding
- 11 DNA machines:
- A are science fiction.
 - B respond to their environment chemically.
 - C are found in genetically modified insulin.
 - D serve no useful purpose.
- 12 Nanotubes:
- A are one atom thick sheets of graphene.
 - B can be made from carbon, silicon and other materials.
 - C are a graphene tube.
 - D are very conductive tubes.

13 Protein tertiary structures:

- A** involve the R groups forming bonds.
- B** occur when the four bases complete their bonding.
- C** occur with several protein structures bonding together.
- D** are a folded sheet of protein material.

14 Which of the following is not found in amino acids?

- A** An amine group
- B** A carboxylic acid group
- C** An alcohol group
- D** An alpha carbon

15 What is reflux?

- A** A technique of heating liquids without losing any ingredients
- B** A process of breaking down reactants into products
- C** A process of breaking down a product into reactants
- D** A process used to produce one product

Section B Short answer (15 marks)

Section B consists of five questions. Write your answers in the spaces provided. You are advised to spend 20 minutes on this section.

1 a Describe the top-down and bottom-up methods of molecular manufacture.

b Name and describe one example of molecular manufacture found in nature.

- c** Give two reasons why artificial molecular manufacture is important.

(1 + 1 + 1 = 3 marks)

- 2** Describe the role of ribosomes in protein manufacture.

(3 marks)

- 3** Silver has a special property – it can kill bacteria. Silver nanoparticles are added to clothing material to reduce smells from bacteria.

- a** Describe how nanoparticles help the effect of silver on bacteria in clothing.

- b** How is the size of nanoparticles different from other particles? How does this help the silver embedded clothing?

(2 + 1 = 3 marks)

- 4** Why is reflux used to produce an ester? Describe the process and how it preserves the product being produced. Use an annotated diagram in your answer.

(3 marks)

- 5** Whenever carbon has four different groups attached to it, their arrangement can be altered uniquely. Show an example of this and describe this importance in how a molecule can affect the human body.

(3 marks)